

Problem set 4: International Trade I (comparative advantages & Surplus)

A. Comparative advantage

1. Explain why companies in developed countries often choose to outsource their call centers to developing countries (in your answer, please refer to Ricardo's Comparative Advantages Theorem).
2. Country A can produce either 2 oranges or 6 lemons. Country B can produce either 2 oranges or 10 lemons. The two countries can trade.
 - a. Which country should produce which product?
 - b. What is the range of prices in which the countries will trade?
 - c. Which of the following options is either inefficient or infeasible?
 - i. Country A produces only oranges and imports 120 lemons in exchange for 30 oranges.
 - ii. Country A produces only oranges and imports 140 lemons in exchange for 40 oranges.
 - iii. Country B produces only lemons and exports 100 lemons in exchange for 25 oranges.
 - iv. Country B produces only lemons and imports 350 oranges in exchange for 100 lemons.
3. Suppose that Germany and Greece that can produce cars and/or airplanes, with the attributes specified in the table below. Assume that there is perfect competition in both markets.
 - a. Show that with these number, Germany *will not export* cars to Greece.
 - b. The german government wants to *decrease* the hourly wage for workers who are producing cars (without changing the wage for producing airplanes). How much would the new wage for workers producing cars have to be in order for Germany to start exporting cars to Greece (assume here that productivity stays the same after the change)?

	Germany	Greece
Cars		
Hourly wage	10	5
Hours worked	1	5
Unit costs	10	25
Airplanes		
Hourly wage	10	5
Hours worked	2	10
Unit costs	20	50

B.Surplus with trade

1. Suppose that a competitive market opens up for trade and the world price is lower than the existing market price, so that there is import. What happens then to the consumer surplus? What happens to the (local) supplier's surplus?
2. When a competitive market opens up for trade and the world price is higher than the existing market price, so that there is export. What happens to the consumer surplus? What happens to the (local) supplier's surplus?
3. Explain why a Taarif protects local suppliers.